

A Technical Introduction to Uncertainty quantification

Section 1

$$R_m((x, \theta), (x', \theta')) = \exp \left\{ - \sum_{k=1}^d \omega_k m(x_k - x'_k)^2 \right\} \exp \left\{ - \sum_{k=1}^r \omega_k m(\theta_k - \theta'_k)^2 \right\}.$$

$$\{\displaystyle R^{\{m\}}(\mathbf{x}, \mathbf{\theta}) = \exp \left\{ - \sum_{k=1}^d \omega_k m(x_k - x'_k)^2 \right\} \exp \left\{ - \sum_{k=1}^r \omega_k m(\theta_k - \theta'_k)^2 \right\}.$$

Section 2

$$y_e(x) = y_m(x, \theta^*) + \delta(x) + \varepsilon$$

$$y^e(\mathbf{x}) = y^m(\mathbf{x}, \mathbf{\theta}^*) + \delta(\mathbf{x}) + \varepsilon$$

Section 3

Simulation-based methods: Monte Carlo simulations, importance sampling, adaptive sampling, etc. General surrogate-based methods: In a non-intrusive approach, a surrogate model is learnt in order to replace the experiment or the simulation with a cheap and fast approximation. Surrogate-based methods can also be employed in a fully Bayesian fashion. This approach has proven particularly powerful when the cost of sampling, e.g. computationally expensive simulations, is prohibitively high. Local expansion-based methods: Taylor series, perturbation method, etc. These methods have advantages when dealing with relatively small input variability and outputs that don't express high nonlinearity. These linear or linearized methods are detailed in the article Uncertainty propagation. Functional expansion-based methods: Neumann expansion, orthogonal or Karhunen–Loeve expansions (KLE), with polynomial chaos expansion (PCE) and wavelet expansions as special cases. Most probable point (MPP)-based methods: first-order reliability method (FORM) and second-order reliability method (SORM). Numerical integration-based methods: Full factorial numerical integration (FFNI) and dimension reduction (DR). For non-probabilistic approaches, interval analysis, Fuzzy theory, Possibility theory and evidence theory are among the most widely used. The probabilistic approach is considered as the most rigorous approach to uncertainty analysis in engineering design due to its consistency with the theory of decision analysis. Its cornerstone is the calculation of probability density functions for sampling statistics. This can be performed rigorously for random variables that are obtainable as transformations of Gaussian variables, leading to exact confidence intervals.

Section 4

Uncertainty propagation is the quantification of uncertainties in system output(s) propagated from uncertain inputs. It focuses on the influence on the outputs from the parametric variability listed in the sources of uncertainty. The targets of uncertainty propagation analysis can be:

Section 5

$$\delta(x) \sim GP(h\delta(\cdot)T\beta\delta, \sigma\delta^2R\delta(\cdot, \cdot))$$

$$\delta(\mathbf{x}) \sim GP(h\delta(\cdot)T\beta\delta, \sigma\delta^2R\delta(\cdot, \cdot))$$

Section 6

$$y_m(x, \theta) \sim GP(hm(\cdot)T\beta m, \sigma m^2Rm(\cdot, \cdot))$$

$$y^m(\mathbf{x}, \mathbf{\theta}) \sim GP(hm(\cdot)T\beta m, \sigma m^2Rm(\cdot, \cdot))$$

Section 7

Modular approach An approach to inverse uncertainty quantification is the modular Bayesian approach. The modular Bayesian approach derives its name from its four-module procedure. Apart from the current available data, a prior distribution of unknown parameters should be assigned.

Section 8

Full approach Fully Bayesian approach requires that not only the priors for unknown parameters θ but also the priors for the other hyperparameters ϕ should be assigned. It follows the following steps:

Section 9

$$R_\delta(x, x') = \exp \left\{ - \sum_{k=1}^d \omega_k \delta(x_k - x'_k)^2 \right\}.$$

$$R^{\{\delta\}}(\mathbf{x}, \mathbf{x}') = \exp \left\{ - \sum_{k=1}^d \omega_k \delta(x_k - x'_k)^2 \right\}.$$

Section 10

Types of problems There are two major types of problems in uncertainty quantification: one is the forward propagation of uncertainty (where the various sources of uncertainty are propagated through the model to predict the overall uncertainty in the system response) and the other is the inverse assessment of model uncertainty and parameter uncertainty (where the model parameters are calibrated simultaneously using test data). There has been a proliferation of research on the former problem and a majority of uncertainty analysis techniques were developed for it. On the other hand, the latter problem is drawing increasing attention in the engineering design community, since uncertainty quantification of a model and the

subsequent predictions of the true system response(s) are of great interest in designing robust systems.

Section 11

Bayesian Several methodologies for inverse uncertainty quantification exist under the Bayesian framework. The most complicated direction is to aim at solving problems with both bias correction and parameter calibration. The challenges of such problems include not only the influences from model inadequacy and parameter uncertainty, but also the lack of data from both computer simulations and experiments. A common situation is that the input settings are not the same over experiments and simulations. Another common situation is that parameters derived from experiments are input to simulations. For computationally expensive simulations, then often a surrogate model, e.g. a Gaussian process or a Polynomial Chaos Expansion, is necessary, defining an inverse problem for finding the surrogate model that best approximates the simulations.

Section 12

d is the dimension of input variables, and r is the dimension of unknown parameters. While $\mathbf{h}^m(\cdot)$ is pre-defined, $\{\beta_m, \sigma_m, \omega_k^m, k = 1, \dots, d+r\}$, known as hyperparameters of the GP model, need to be estimated via maximum likelihood estimation (MLE). This module can be considered as a generalized kriging method.

Section 13

where $y^e(\mathbf{x})$ denotes the experimental measurements as a function of several input variables $\mathbf{x}$, $y^m(\mathbf{x})$ denotes the computer model (mathematical model) response, $\delta(\mathbf{x})$ denotes the additive discrepancy function (aka bias function), and ε denotes the experimental uncertainty. The objective is to estimate the discrepancy function $\delta(\mathbf{x})$, and as a by-product, the resulting updated model is $y^m(\mathbf{x}) + \delta(\mathbf{x})$. A prediction confidence interval is provided with the updated model as the quantification of the uncertainty.